

Logarithmic Equations

$$\log_3(x-1) = 2$$

$$x-1 = 3^2$$

$$x = 10$$

$$\log_2(x) + \log_2(x-3) = 3$$

$$\log_2(x(x-3)) = 3$$

$$\log_2(x^2 - 3x) = 3$$

$$x^2 - 3x = 2^3$$

$$x^2 - 3x - 8 = 0$$

$$x = \frac{3 \pm \sqrt{41}}{2}$$

$$D.F: \begin{matrix} x-1 > 0 \\ x > 1 \end{matrix}$$

$$\Rightarrow \boxed{x = \frac{3 + \sqrt{41}}{2}}$$

$$2^{3x-1} = 7$$

$$\Rightarrow \log_2(7) = 3x-1$$

$$\frac{\ln 7}{\ln 2} = 3x-1$$

$$\ln 7 = (3x-1) \ln 2$$

$$\ln 7 = \ln(2)^{3x-1}$$

$$e^{\ln 7} = e^{\ln(2)^{3x-1}}$$

$$7 = 2^{3x-1}$$

$$\Rightarrow \frac{\ln 7}{\ln 2} + 1 = 3x$$

$$x = \frac{\frac{\ln 7}{\ln 2} + 1}{3}$$

$$\text{Using cal: } x \approx 1.269$$

$$2^x - 3 \cdot 2^{x-1} = 0$$

$$2^x = +3 \cdot 2^{x-1}$$

$$\text{Let } u = 2^x \Rightarrow u - 3u^{-1} = 0$$

$$3u^{-1} = u$$

$$u^{-1} = \frac{u}{3}$$

$$2^{x-1} = \frac{2^x}{3}$$